AHKAR HEIN

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SUMMARY

Software developer with an M.S. in Software Development and full-stack experience building real applications in MERN, Python/FastAPI, and PHP/MySQL, using AI-assisted tools (Claude, GitHub Copilot) to write, debug, and review code. Currently a Production Technician at PsiQuantum, performing precision optical alignment on photonic chips and monitoring live measurement data in a cleanroom environment. Aiming to grow into Automation Software or Industrial Engineering, combining coding skills with hands-on manufacturing experience to build tools that make production faster and more reliable.

TECHNICAL SKILLS

Languages: JavaScript, Python, SQL, HTML, CSS, Java, C++, PHP, .NET

Web & Backend: Node.js, Express.js, React.js, FastAPI, REST APIs, GraphQL, JWT, Redux, MongoDB, Mongoose ODM, PostgreSQL, Sequelize ORM

Integrations & Tools: Git/GitHub, AI-assisted development (Claude, GitHub Copilot), Twilio, Cloudinary, Firebase, Socket.io, Docker, AWS, Tailwind CSS, Bootstrap

Hardware & Lab: Active optical alignment, edge coupling, fiber handling, optical power & spectrum measurement, microscope inspection, MES, cleanroom/ESD, wiring schematics

Practices: Agile & Scrum, Version Control & Code Review, Documentation & Structured Workflows, Cross-functional Collaboration

EXPERIENCE

Production Technician — PsiQuantum, San Jose, CA Sep 2026 – Present

- Perform active optical alignment and edge coupling between fiber array units (FAUs) and photonic chips, and chip-to-chip, using laser sources and precision positioning to maximize optical coupling power.
- Assemble, inspect, and clean semiconductor dies, chips, and FAUs under high-magnification cameras and microscopes, following cleanroom, ESD, and material-handling requirements.
- Monitor live spectrum measurements and real-time power plots to verify coupling performance; troubleshoot equipment, software, and automation interfaces.
- Track builds, materials, and WIP through MES transactions and deliver daily shift pass-downs to Operations and Engineering on status, issues, and process improvements.

Electro-Mechanical Assembly Operator — Siemens Energy, Pomona, CA Nov 2024 – Jun 2026

- Assembled and tested electro-mechanical components against production and quality standards, documenting results in a structured, repeatable process.
- Partnered with engineers to troubleshoot hardware issues through systematic root-cause analysis.
- Read wiring schematics and applied component testing procedures to support reliable assembly.

Full-Stack Web Developer — Hyatt Knitwear, Yangon, Myanmar Oct 2020 – Feb 2021

- Built a barcode and production-tracking web app for a knitwear manufacturer (PHP, MySQL, jQuery/AJAX, Bootstrap), covering order and plan import, color/size matrices, price approval, and employee production records.
- Wrote 20+ MySQL stored procedures to generate barcodes by plan, order, color, and size, manage per-user import tables, and power date, plan, and employee reports.
- Implemented role-based access for Admin, Manager, Supervisor, and HR users, session timeouts, and one-click CSV/PDF report exports.

PORTFOLIO PROJECTS

Sentry — Neighborhood Safety Platform 2025 – 2026

MERN Stack, Python FastAPI, Socket.io, WebRTC, Twilio, Cloudinary, Firebase

- Built and deployed a full-stack, geo-fenced community safety app (React/Vite, Node.js/Express, MongoDB) with a Python FastAPI service, real-time chat, and an emergency panic-button alert flow.
- Integrated Twilio (SMS alerts), Cloudinary (media), Firebase (push notifications), and Socket.io/WebRTC (real-time communication) into one cohesive platform.
- Used AI-assisted tools to speed up coding and debugging, reviewing and correcting AI-generated code before shipping.

Loud-Mouth Social Media App 2023

MongoDB, Express.js, React.js, Node.js, GraphQL, Mongoose ODM, JWT

- Developed a full-stack social app with GraphQL APIs for posts, reactions, and user interactions, backed by JWT authentication.

EDUCATION

Full Stack Web Development Bootcamp — UC Berkeley Extension, CA Jun 2023 – Sep 2023

M.S. in Software Development — Coventry University, Coventry, UK Sep 2019 – Sep 2020

B.S. Business Information Systems / HND Computing — Middlesex University, London, UK Sep 2016 – Jun 2019